Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, DECEMBER 2024

ENERGY AUDITING [ELE 4082]

REVISED CREDIT SYSTEM

Time: 3 Hours **26 December 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. A manufacturing industry plans to improve its energy performance under PAT through implementation of an energy conservation scheme. After implementation, calculate the Plant Energy Performance (PEP) with 2015-16 as the reference year. What is your inference? Given that:

The current year (2016-17) Annual Production – 28,750 Tons, Current year (2016-17) Annual Energy Consumption– 23,834 MWh, Reference year (2015-16) production - 34,000 Tons, Reference year (2015-16) Energy consumption - 27,200 MWh. ***(03)***

1B. Explain **"Availability Based Tariff (ABT)"**, with the help of a suitable example. ***(03)***

1C. An ESCO company is required to invest in a waste heat recovery project, which is expected to yield an annual saving of Rs.10,00,000 and the life of the equipment is 7 years. If the ESCO expects 30% IRR on this project, calculate the investment required to be made. ***(04)***

2A. What is electrical tariff? What are the factors that are involved in deciding final cost of purchased electricity? ***(03)***

2B. The cost characteristic of two units in a plant are:
$C_1 = 0.4\ P_1^2 + 160\ P_1 + K_1$ Rs. /h
$C_2 = 0.45\ P_2^2 + 120\ P_2 + K2$ Rs. / h
where P_1 and P_2 are power output in MW. Find the optimum load allocation between the two units, when the total load is 162.5 MW. What will be the daily loss if the units are loaded equally? ***(03)***

2C. Explain any **FOUR** objectives of **Demand Side Management**. ***(04)***

3A. A 40W Fluorescent lamp consisting of magnetic ballast is powered from a 230V, 50Hz supply, it is desired to improve its power factor of this set-up from 0.3 lag to 0.7 lag.

- What is the value of the capacitor in Micro-Farads that is required for PF improvement?
- What is difference energy consumption before and after PF improvement? ***(03)***

3B. Draw a neat, labelled drawing of "**Heat Transfer loops in Refrigeration Systems**". What is the fundamental energy engineering operation in these loops? ***(03)***

3C. Two main areas of an industrial plant have the following lighting systems:

Area A: 50 x 400W High Pressure Sodium (HPSV) single lamp luminaires.
Area B: 35 x 400W Mercury Vapour (HPMV) single lamp luminaires.

In Area A and Area B, the measured illuminance during daylight hours (12 hours) without artificial light was found to be adequate.
In Area B it was noted that 8 of the HPMV fixtures are redundant.
Calculate the annual potential energy cost savings from switching off unnecessary lights and from disconnecting redundant luminaires? Assuming plant operating hours as 8760 hours and cost of energy at Rs 6 /unit ***(04)***

4A. With the help of an equation explain **principle of combustion**. What is the output when there is incomplete, good, and perfect combustion? ***(03)***

4B. A liquid waste stream has a flow rate of 3.5 kg/s and a temperature of 70° C with a specific heat capacity of 4190 J/kgK. Heat recovered from the hot waste stream is used to pre-heat boiler make-up water. The flow rate of the make-up water is 2 kg/s, its temperature is 10oC and its specific heat capacity is 4190 J/kg/K. The overall heat transfer coefficient of the heat exchanger is 800 W/m^2K. If a make-up water exit temperature of $50^{o}C$ is required, and assuming that there is no heat losses from the exchanger, determine

1) The heat transfer rate
2) The exit temperature of the effluent and
3) The area of the heat exchanger required. ***(03)***

4C. Draw a neat, labeled sketch of a "**Gas turbine cogeneration systems**". ***(04)***

5A. A solar photovoltaic power plant is installed in roof top area of a building having dimension of 9 m x 10m. The panel dimensions are 1.5 m x 1.5 m and gives an output of 350 Watts. If solar insolation (sunlight falling on the panel) is 1,000 W/m^2, calculate the panel conversion efficiency? ***(03)***

5B. Draw a neat sketch of a Solar Based Thermal power plant and label its parts. What advantages does Solar Thermal power plant offer in comparison with fossil fuel based thermal power plant. ***(03)***

5C. A load which is situated near a river requires continuous power of 600 kW all through the year. Head available is 25m. The water flow in a year is.

i) 5 m^3/s for 4 months
ii) 3 m^3/s for 4 months and
iii) 6 m^3/s for 4 months.

Assuming the overall plant efficiency as 80%,

Find the following:

i. the rating of standby unit required to supply the deficit power if there is no reservoir and water is utilized as and when it flows.
i) with reservoir, what will be the excess power available? ***(04)***